# Object Oriented Design and Implementation of a Military Aviation Spare Parts Inventory System

**First Semester 2026-27**
Project Oct/Nov 2026 | Document D2

SS G514 OBJECT ORIENTED ANALYSIS & DESIGN

**Area: 40, Defence**

| **STUDENT DETAILS** | |
|---|---|
| **Name** | Milap Chaudhari |
| **ID No.** | 2025H1030207P |
| **Course** | SS G514 Object Oriented Analysis & Design |
| **Paper selected** | The efficiency of using a tailored inventory management system in the military aviation industry |

## Programming language and requirements

| Item | Requirement |
|---|---|
| Language | Java 17 source compatibility; source compiles with JDK 17 or later. |
| Build | Apache Maven 3.8+; JUnit 5 dependency for tests. |
| Hardware | Any standard computer capable of running a Java 17+ JDK. No special hardware is needed. |
| Other software | Visual Studio Code with Extension Pack for Java is optional. No database server or third-party runtime package is required. |
| Data | Local CSV files are created in inventory-data/ when the program first runs. |
| Network | Maven may need network access the first time to download JUnit and build plugins. |

## Build, test, and run

1. Open the project folder in VS Code or a terminal.
2. Select JDK 17 or later for Maven. Make sure the JDK bin directory is on PATH before building.
3. Run mvn test. The submitted test run completed 6 tests with 0 failures.
4. Run java -cp target/classes h2025H1030207P_D2 from the project root.

## Demo accounts and permissions

| Login | Role | Main actions |
|---|---|---|
| keeper | Store keeper | Add/update parts, receive and issue stock, repair returns, reports. |
| tech | Technician | Raise requests, declare AOG, fit issued parts, reports. |
| planner | Planner | Select policy, review and approve proposals, approve non-critical requests. |
| commander | Commander | Approve critical-part requisitions, view reports and AOG. |

## Repeatable AOG demonstration

1. Log in as planner, choose menu option 10, select minmax, then log out. Set the policy before the request so its proposal uses the chosen strategy.

2. Log in as tech and choose option 2. Enter tail AC-01, priority aog, one line for ROT-220 quantity 1, and grounded y. The sample starts with no ROT-220 stock.

3. Log in as planner. Option 11 shows proposals; option 12 approves ROT-220.

4. Log in as keeper. Option 17 receives the approved proposal. Option 1 lists the generated serial.

5. Log in as commander. Option 3 approves the critical requisition. Log in as keeper and option 5 issues it.

6. Log in as tech. Use option 1 to find the serial, option 6 to fit it, option 14 for the AOG report, and option 13 as keeper for the stock report.

7. Choose option 19 to compare fixed-level and minimum-maximum policy results on synthetic demand.

## Sample files

The submission bundle includes h2025H1030207P_D2.ip1 with one input response per console prompt and h2025H1030207P_D2.op1 with the captured output. All catalogue and aircraft data are fictional. To restore fresh initial state, close the program and remove its inventory-data folder.